

Ordinary Least Squares

POLS 602

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Review

1. What does the sum of squared errors (SSE) measure?
2. What does the dot product between two vectors capture?

Today

- How do we figure out which line is the line of best fit?
- Understand the properties of the OLS estimator
- How to think about models and variables geometrically

The Goal

What Are We Looking For?

We want the line

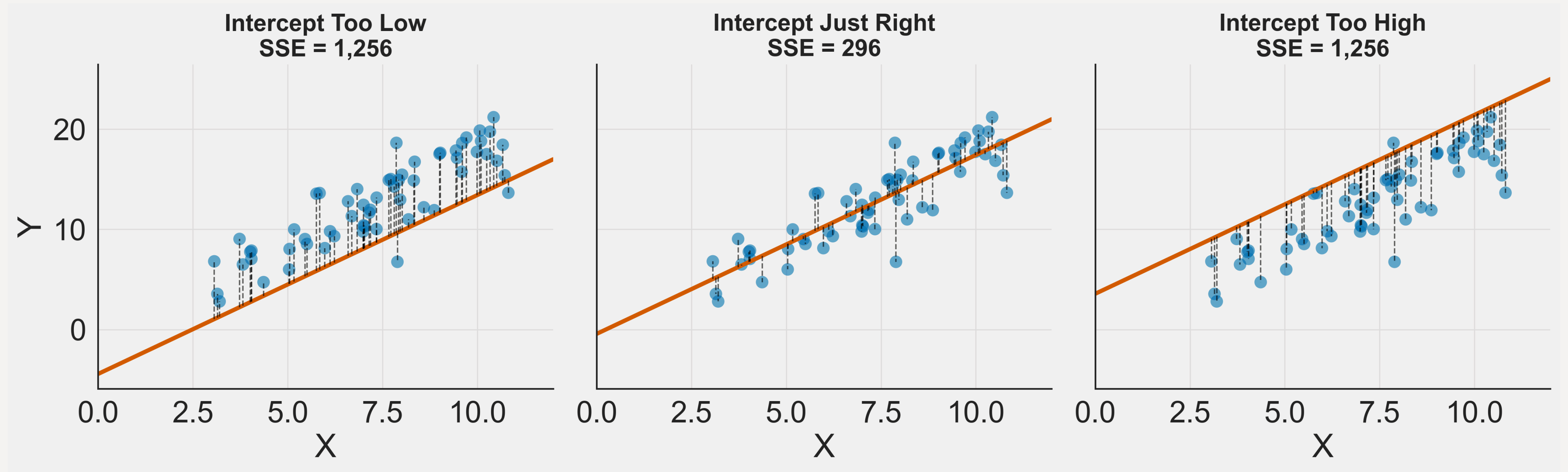
$$\widehat{Y}_i = \widehat{\alpha} + \widehat{\beta}X_i$$

that makes the sum of squared errors as small as possible:

$$SSE = \sum_{i=1}^n \widehat{\epsilon}_i^2 = \sum_{i=1}^n (Y_i - \widehat{Y}_i)^2$$

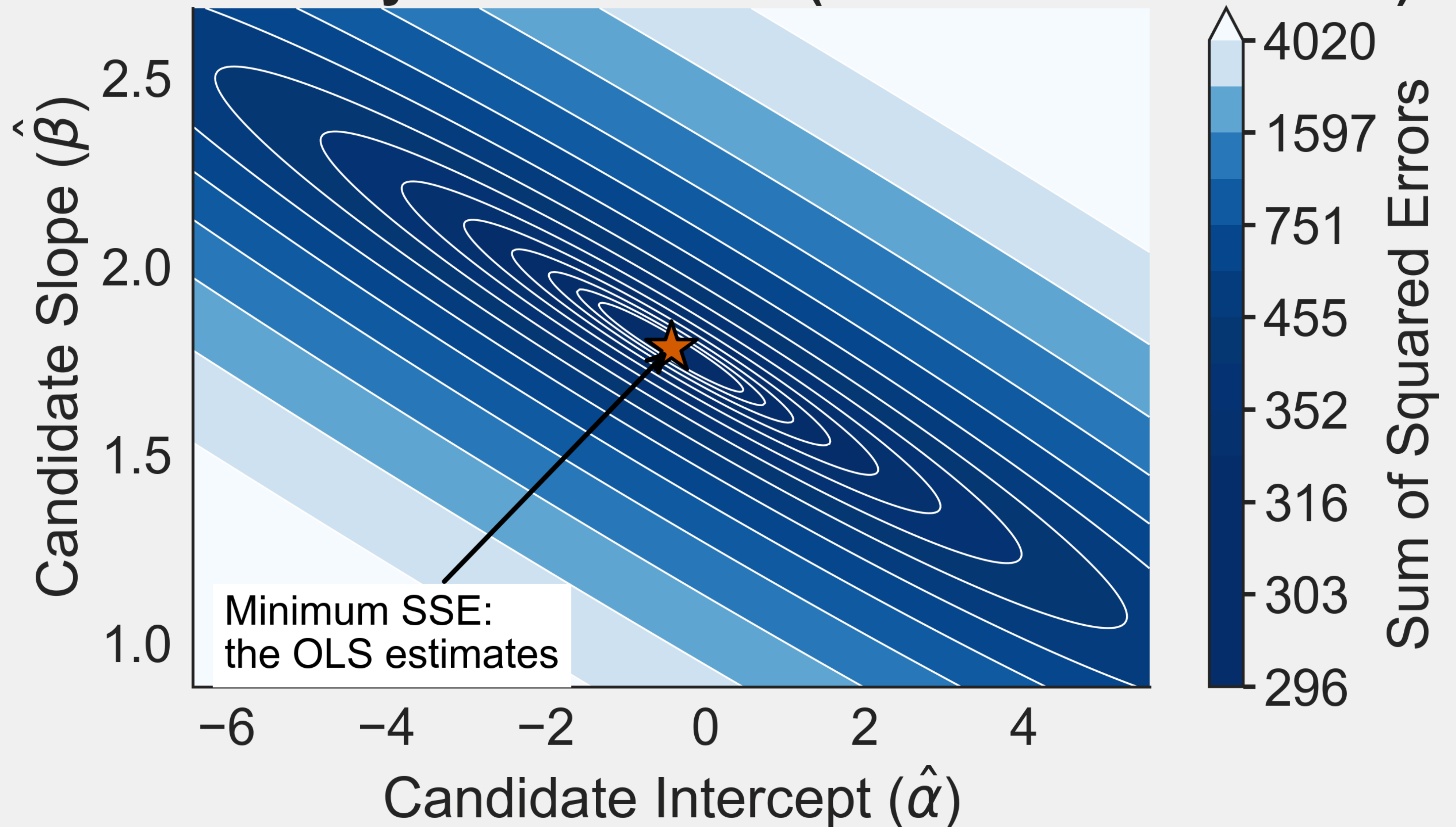
Ordinary Least Squares (OLS): the values of $\widehat{\alpha}$ and $\widehat{\beta}$ that give the *least squared* errors.

SSE Depends on the Line We Choose



SSE Depends on Both Parameters

SSE Across Every Possible Line (Darker = Smaller SSE)



Our Strategy

1. Write SSE as a function of $\hat{\alpha}$ and $\hat{\beta}$
2. Use calculus to find the bottom of the bowl
3. Solve for $\hat{\alpha}$
4. Use that to solve for $\hat{\beta}$

Calculus Refresher

What Is a Derivative?

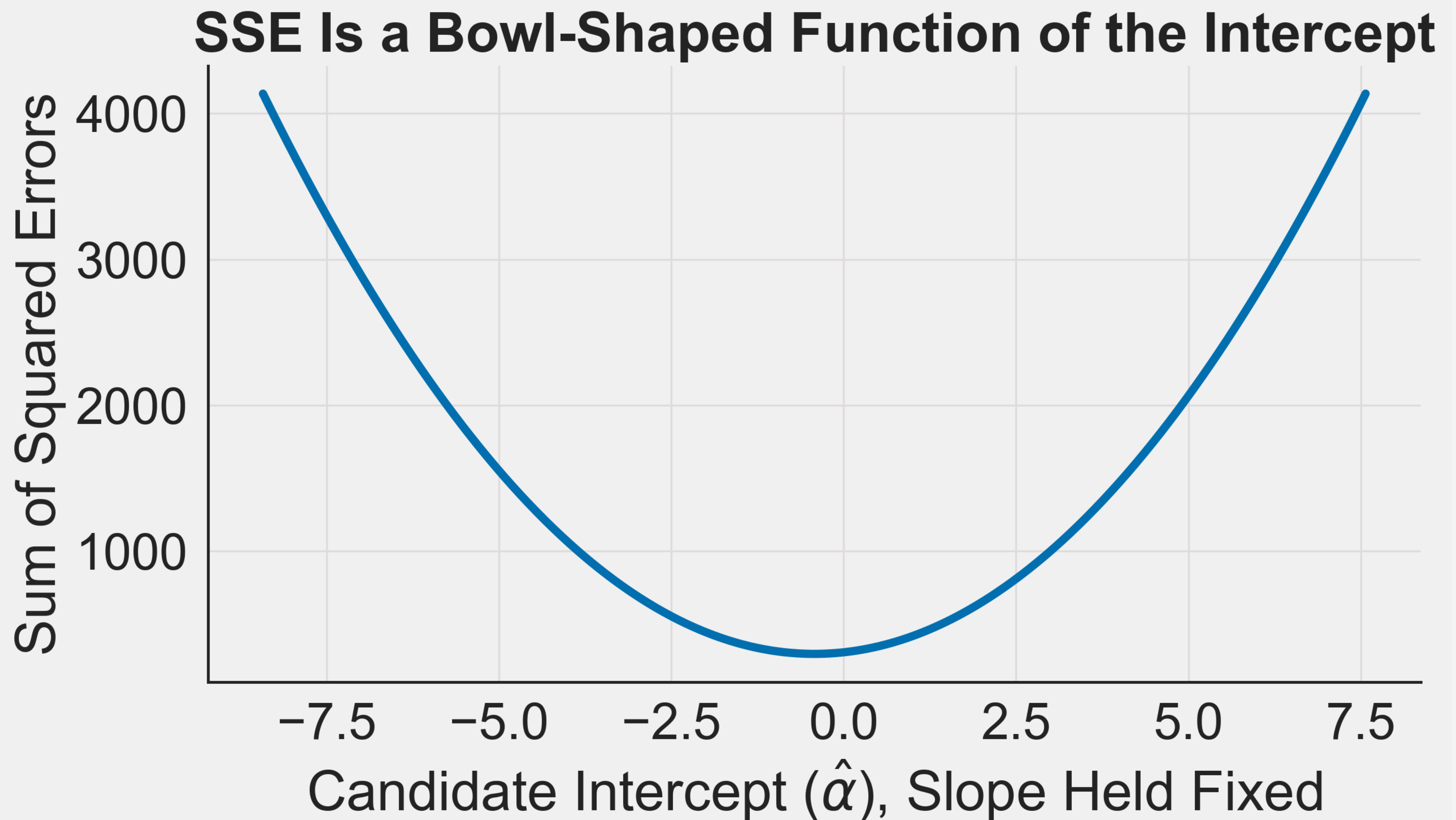
The derivative of a function tells us its **slope at a point**.

In plain English: *if I nudge the input up by a tiny amount, how much does the output change, and in which direction?*

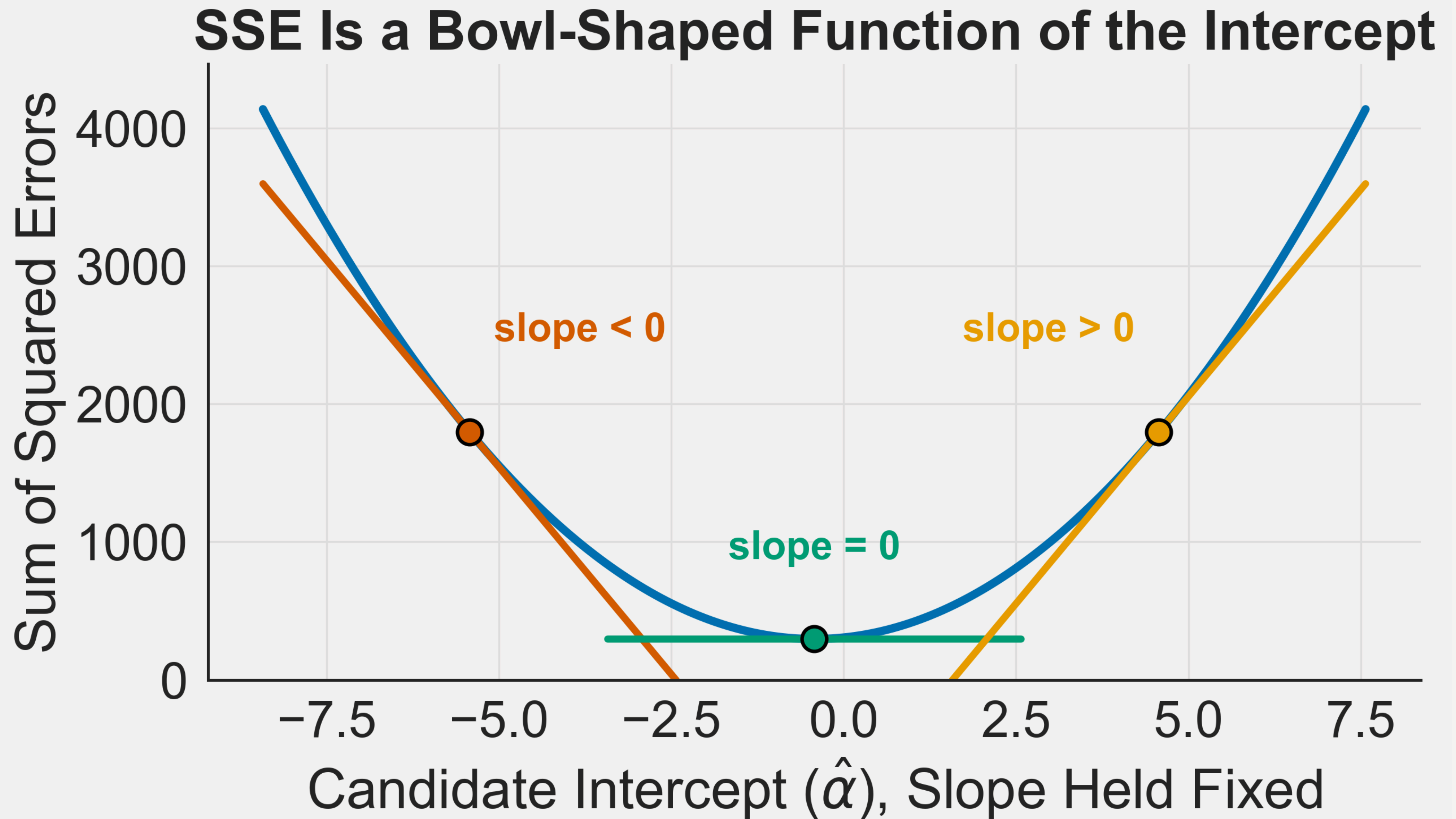
Here, the function is SSE and the input is our choice of $\hat{\alpha}$:

If I nudge the intercept up a tiny bit, does SSE go up or down, and how fast?

SSE as a Function of the Intercept



The Slope of the Bowl



Two Parameters: Partial Derivatives

SSE depends on both $\hat{\alpha}$ and $\hat{\beta}$.

- $\frac{\partial SSE}{\partial \hat{\alpha}}$: how SSE changes as we nudge $\hat{\alpha}$, **holding $\hat{\beta}$ constant**
- $\frac{\partial SSE}{\partial \hat{\beta}}$: how SSE changes as we nudge $\hat{\beta}$, **holding $\hat{\alpha}$ constant**

At the bottom of the bowl, the ground is flat in *every* direction, so **both** partial derivatives must equal zero.

Derivative Rules We Will Use

- **Constant rule:** $\frac{d}{dx} c = 0$
- **Constant multiple rule:** $\frac{d}{dx} cf(x) = c \frac{d}{dx} f(x)$
- **Sum rule:** $\frac{d}{dx} [f(x) + g(x)] = \frac{d}{dx} f(x) + \frac{d}{dx} g(x)$
- **Power rule:** $\frac{d}{dx} x^2 = 2x$
- **Chain rule:** $\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$

The Chain Rule

When a function is nested inside another function, differentiate the **outside**, leave the inside alone, then multiply by the derivative of the **inside**.

Example: $\frac{d}{dx} (3x + 1)^2$

- **Outside function**: “something squared”. Its derivative is $2 \times (\text{something})$
- **Inside function**: $3x + 1$. Its derivative is 3

$$\frac{d}{dx} (3x + 1)^2 = 2(3x + 1) \cdot 3 = 6(3x + 1)$$

Setting Up the Problem

The Fitted Line and Its Residuals

$$\widehat{Y}_i = \widehat{\alpha} + \widehat{\beta}X_i$$

$$\widehat{\epsilon}_i = Y_i - \widehat{Y}_i = Y_i - \widehat{\alpha} - \widehat{\beta}X_i$$

SSE as a Function of the Parameters

$$SSE(\hat{\alpha}, \hat{\beta}) = \sum_{i=1}^n (Y_i - \hat{\alpha} - \hat{\beta}X_i)^2$$

- X_i and Y_i are fixed numbers we observed
- $\hat{\alpha}$ and $\hat{\beta}$ are the variables

The Plan

Take both partial derivatives and set each equal to zero:

$$\frac{\partial SSE}{\partial \hat{\alpha}} = 0$$

$$\frac{\partial SSE}{\partial \hat{\beta}} = 0$$

First we solve for $\hat{\alpha}$, then plug that into the equation and solve for $\hat{\beta}$.

Solving for the Intercept

Step 1: Differentiate with Respect to $\hat{\alpha}$

$$\frac{\partial SSE}{\partial \hat{\alpha}} = \frac{\partial}{\partial \hat{\alpha}} \sum_{i=1}^n \left(Y_i - \hat{\alpha} - \hat{\beta} X_i \right)^2$$

Sum rule: the derivative of a sum is the sum of the derivatives, so we can move the derivative inside the $\sum$:

$$= \sum_{i=1}^n \frac{\partial}{\partial \hat{\alpha}} \left(Y_i - \hat{\alpha} - \hat{\beta} X_i \right)^2$$

Chain Rule: Identify the Pieces

$$\frac{\partial}{\partial \hat{\alpha}} \left(Y_i - \hat{\alpha} - \hat{\beta} X_i \right)^2$$

- **Outside function:** something squared
- **Inside function:** $Y_i - \hat{\alpha} - \hat{\beta} X_i$

Chain Rule: Outside Derivative

Power rule on the outside function: the derivative of u^2 is $2u$.
Leave the inside alone.

$$2 (Y_i - \hat{\alpha} - \hat{\beta} X_i)$$

Chain Rule: Inside Derivative

Differentiate $Y_i - \hat{\alpha} - \hat{\beta}X_i$ with respect to $\hat{\alpha}$, term by term:

- Y_i : a constant (data), so the derivative is 0 (*constant rule*)
- $-\hat{\alpha}$: the derivative is -1
- $-\hat{\beta}X_i$: no $\hat{\alpha}$ in it, so it's a constant here, and the derivative is 0 (*constant rule*)

$$\frac{\partial}{\partial \hat{\alpha}} (Y_i - \hat{\alpha} - \hat{\beta}X_i) = -1$$

Chain Rule: Multiply

$$\frac{\partial SSE}{\partial \hat{\alpha}} = \sum_{i=1}^n 2 \left(Y_i - \hat{\alpha} - \hat{\beta} X_i \right) \cdot (-1)$$

Constant multiple rule: 2 and -1 are constants, so pull them out in front of the sum:

$$\frac{\partial SSE}{\partial \hat{\alpha}} = -2 \sum_{i=1}^n \left(Y_i - \hat{\alpha} - \hat{\beta} X_i \right)$$

Step 2: Set It Equal to Zero

We want the bottom of the bowl, where the slope is zero:

$$-2 \sum_{i=1}^n (Y_i - \hat{\alpha} - \hat{\beta} X_i) = 0$$

Divide both sides by -2 :

$$\sum_{i=1}^n (Y_i - \hat{\alpha} - \hat{\beta} X_i) = 0$$

What Does This Say?

$$\sum_{i=1}^n (Y_i - \hat{\alpha} - \hat{\beta}X_i) = 0$$

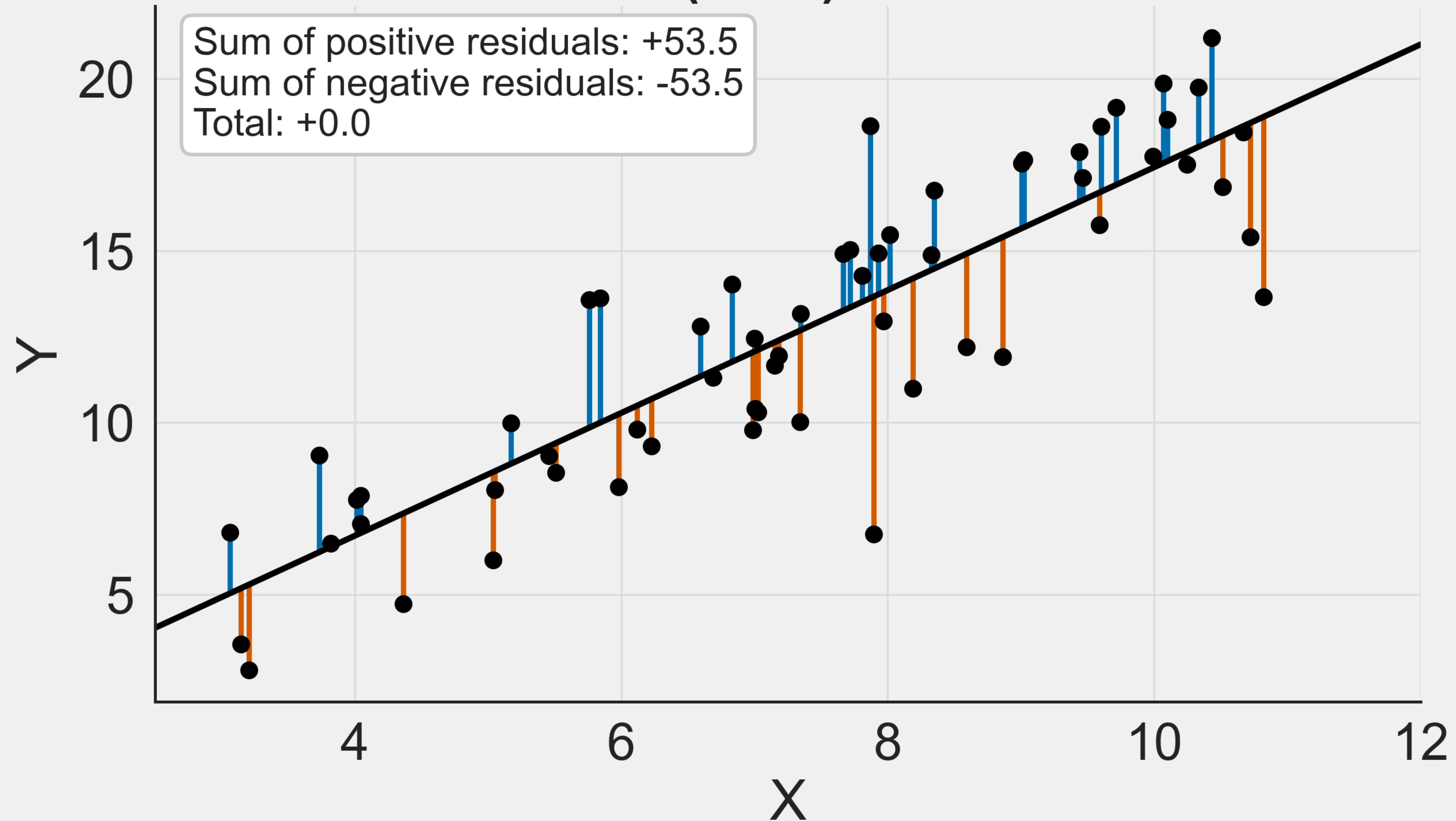
The **blue part** is the residual, $\hat{\epsilon}_i = Y_i - \hat{Y}_i$. So:

The residuals sum to zero.

In plain English: the best line is positioned so that the misses above the line exactly cancel the misses below it.

Residuals Balance Out

Residuals Above the Line (Blue) Balance Those Below (Red)



Step 3: Distribute the Sum

$$\sum_{i=1}^n (Y_i - \hat{\alpha} - \hat{\beta} X_i) = 0$$

$$\sum_{i=1}^n Y_i - \sum_{i=1}^n \hat{\alpha} - \hat{\beta} \sum_{i=1}^n X_i = 0$$

- The sum of a difference is the difference of the sums (*sum rule*)
- $\hat{\beta}$ is a constant, so it comes out of its sum (*constant multiple rule*)

Summing a Constant

$\hat{\alpha}$ doesn't depend on i . Adding the same number n times is just multiplication:

$$\sum_{i=1}^n \hat{\alpha} = \underbrace{\hat{\alpha} + \hat{\alpha} + \cdots + \hat{\alpha}}_{n \text{ times}} = n\hat{\alpha}$$

$$\sum_{i=1}^n Y_i - n\hat{\alpha} - \hat{\beta} \sum_{i=1}^n X_i = 0$$

Step 4: Solve for $\hat{\alpha}$

Move the $n\hat{\alpha}$ term to the other side:

$$n\hat{\alpha} = \sum_{i=1}^n Y_i - \hat{\beta} \sum_{i=1}^n X_i$$

Divide both sides by n :

$$\hat{\alpha} = \frac{1}{n} \sum_{i=1}^n Y_i - \hat{\beta} \frac{1}{n} \sum_{i=1}^n X_i$$

Step 5: Recognize the Means

$$\hat{\alpha} = \frac{1}{n} \sum_{i=1}^n Y_i - \hat{\beta} \frac{1}{n} \sum_{i=1}^n X_i$$

$\bar{Y}$ $\bar{X}$

$$\hat{\alpha} = \bar{Y} - \hat{\beta} \bar{X}$$

Why Does the Intercept Land There?

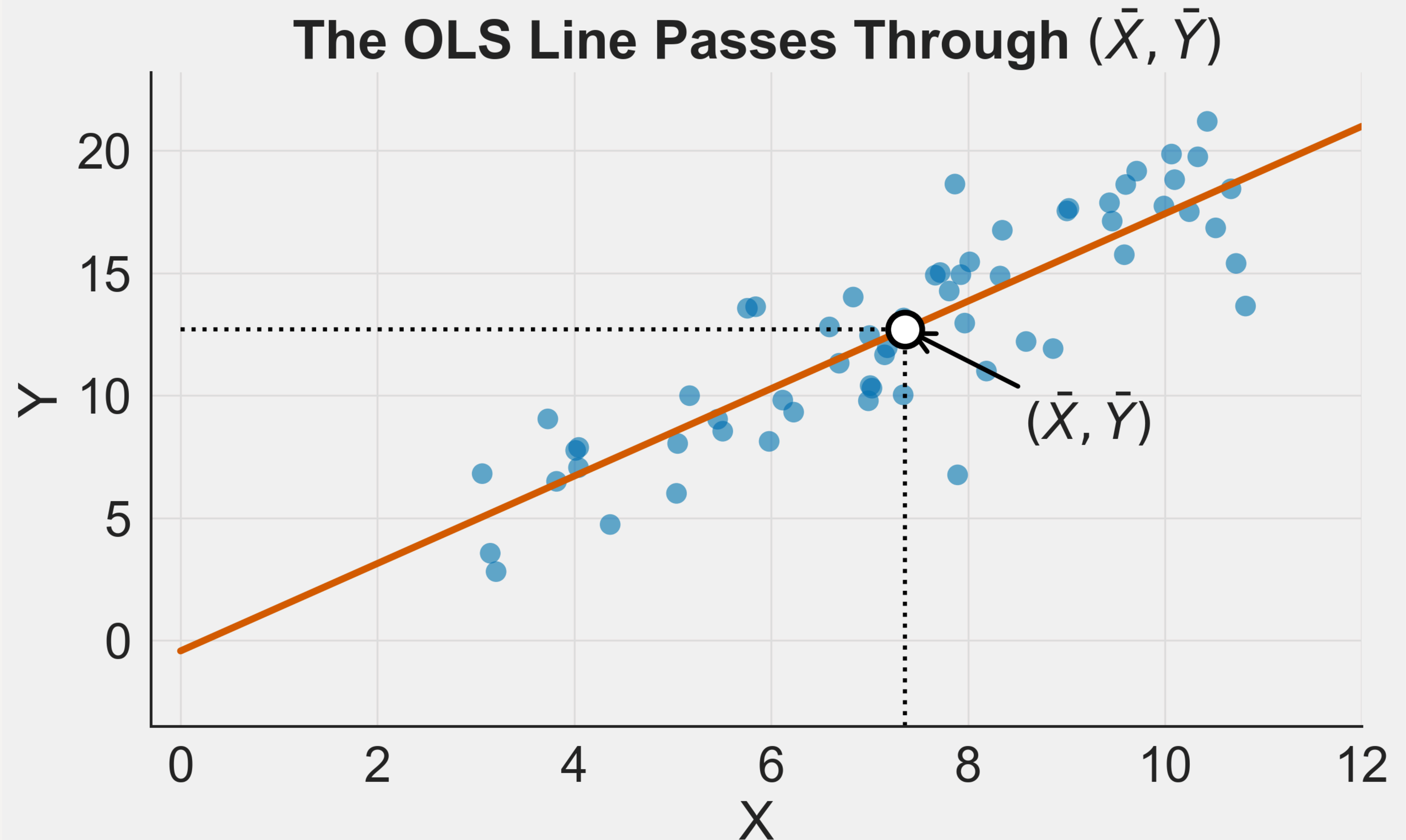
Rearrange $\hat{\alpha} = \bar{Y} - \hat{\beta}\bar{X}$:

$$\bar{Y} = \hat{\alpha} + \hat{\beta}\bar{X}$$

Plug $X = \bar{X}$ into the fitted line and you get $\hat{Y} = \bar{Y}$.

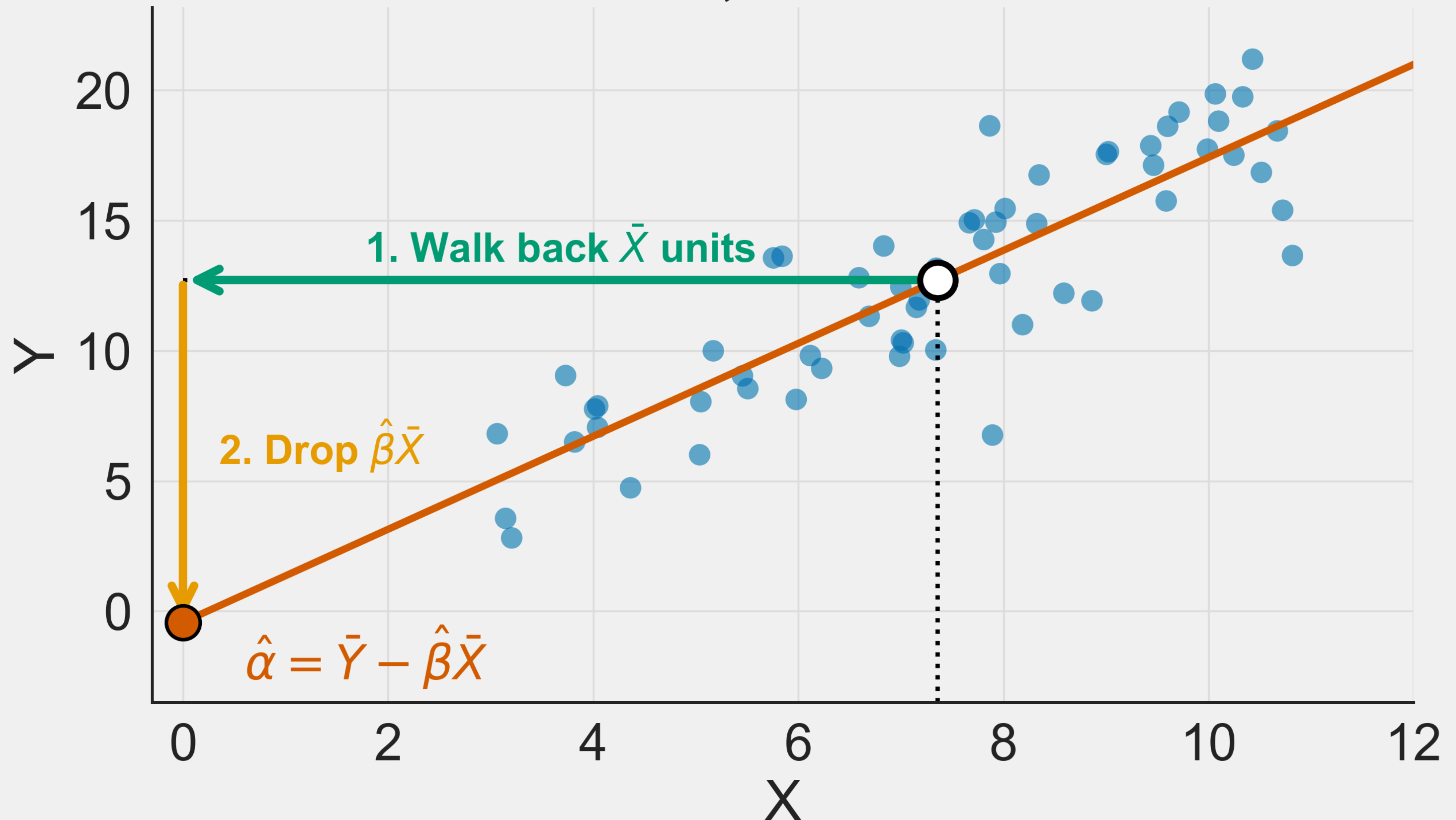
The OLS line always passes through the point $(\bar{X}, \bar{Y})$.

The Line Passes Through the Means



Walking Back to the Intercept

Anchor at the Means, Then Walk Back to $X = 0$



Solving for the Slope

Step 1: Differentiate with Respect to $\hat{\beta}$

$$\frac{\partial SSE}{\partial \hat{\beta}} = \sum_{i=1}^n \frac{\partial}{\partial \hat{\beta}} \left(Y_i - \hat{\alpha} - \hat{\beta} X_i \right)^2$$

Same outside function, so the outside derivative is the same:

$$2 \left(Y_i - \hat{\alpha} - \hat{\beta} X_i \right)$$

Chain Rule: Inside Derivative

Differentiate $Y_i - \hat{\alpha} - \hat{\beta}X_i$ with respect to $\hat{\beta}$, term by term:

- Y_i : a constant, so the derivative is 0 (*constant rule*)
- $-\hat{\alpha}$: no $\hat{\beta}$ in it, so it's a constant here, and the derivative is 0 (*constant rule*)
- $-\hat{\beta}X_i$: X_i is a constant multiplying $\hat{\beta}$, so the derivative is $-X_i$ (*constant multiple rule*)

$$\frac{\partial}{\partial \hat{\beta}} (Y_i - \hat{\alpha} - \hat{\beta}X_i) = -X_i$$

Chain Rule: Multiply and Simplify

$$\frac{\partial SSE}{\partial \hat{\beta}} = \sum_{i=1}^n 2 (Y_i - \hat{\alpha} - \hat{\beta} X_i) \cdot (-X_i)$$

Pull the constant -2 out front (*constant multiple rule*):

$$\frac{\partial SSE}{\partial \hat{\beta}} = -2 \sum_{i=1}^n X_i (Y_i - \hat{\alpha} - \hat{\beta} X_i)$$

Step 2: Set It Equal to Zero

$$-2 \sum_{i=1}^n X_i (Y_i - \hat{\alpha} - \hat{\beta} X_i) = 0$$

Divide both sides by -2 :

$$\sum_{i=1}^n X_i (Y_i - \hat{\alpha} - \hat{\beta} X_i) = 0$$

What Does This Say?

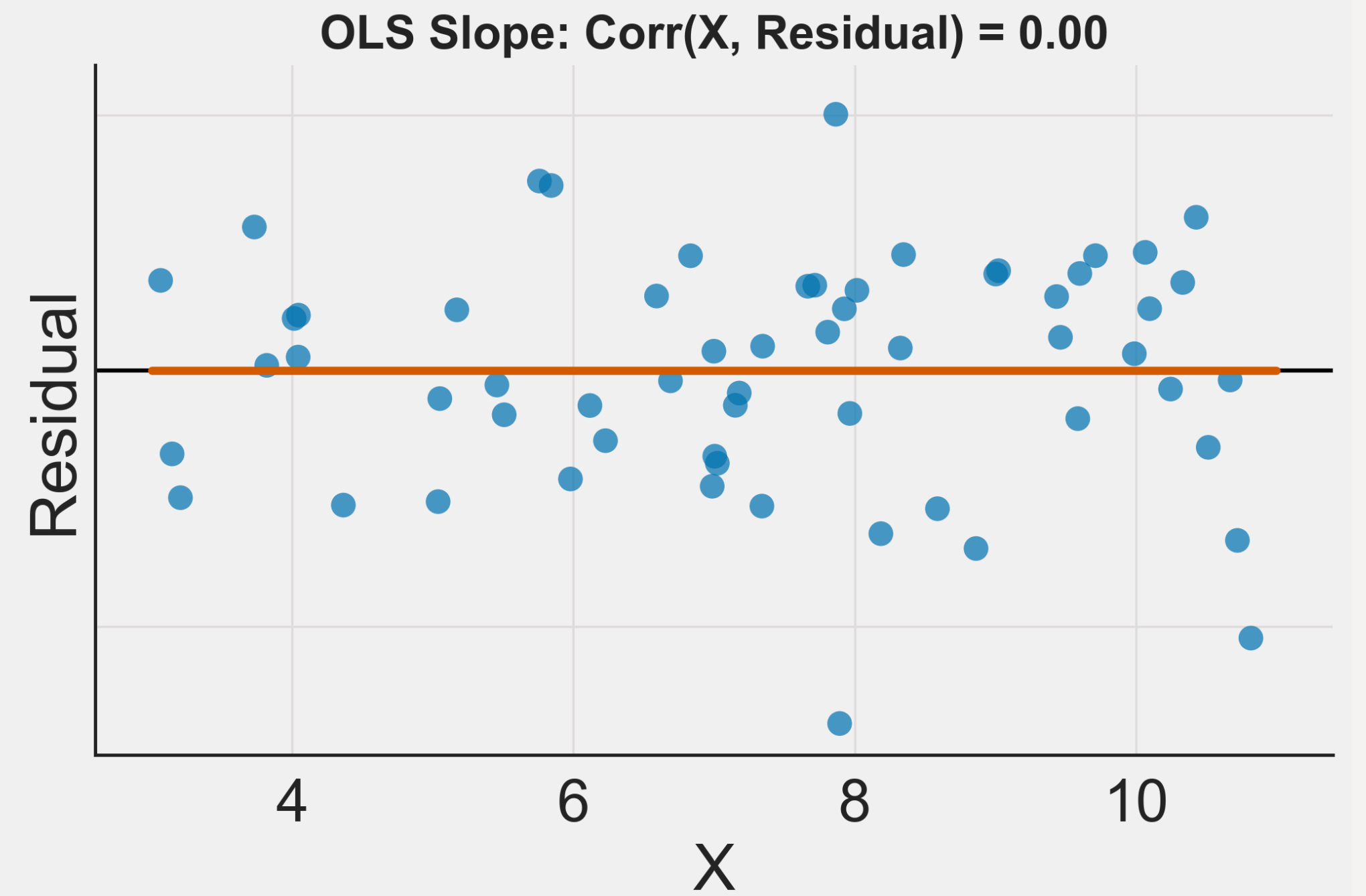
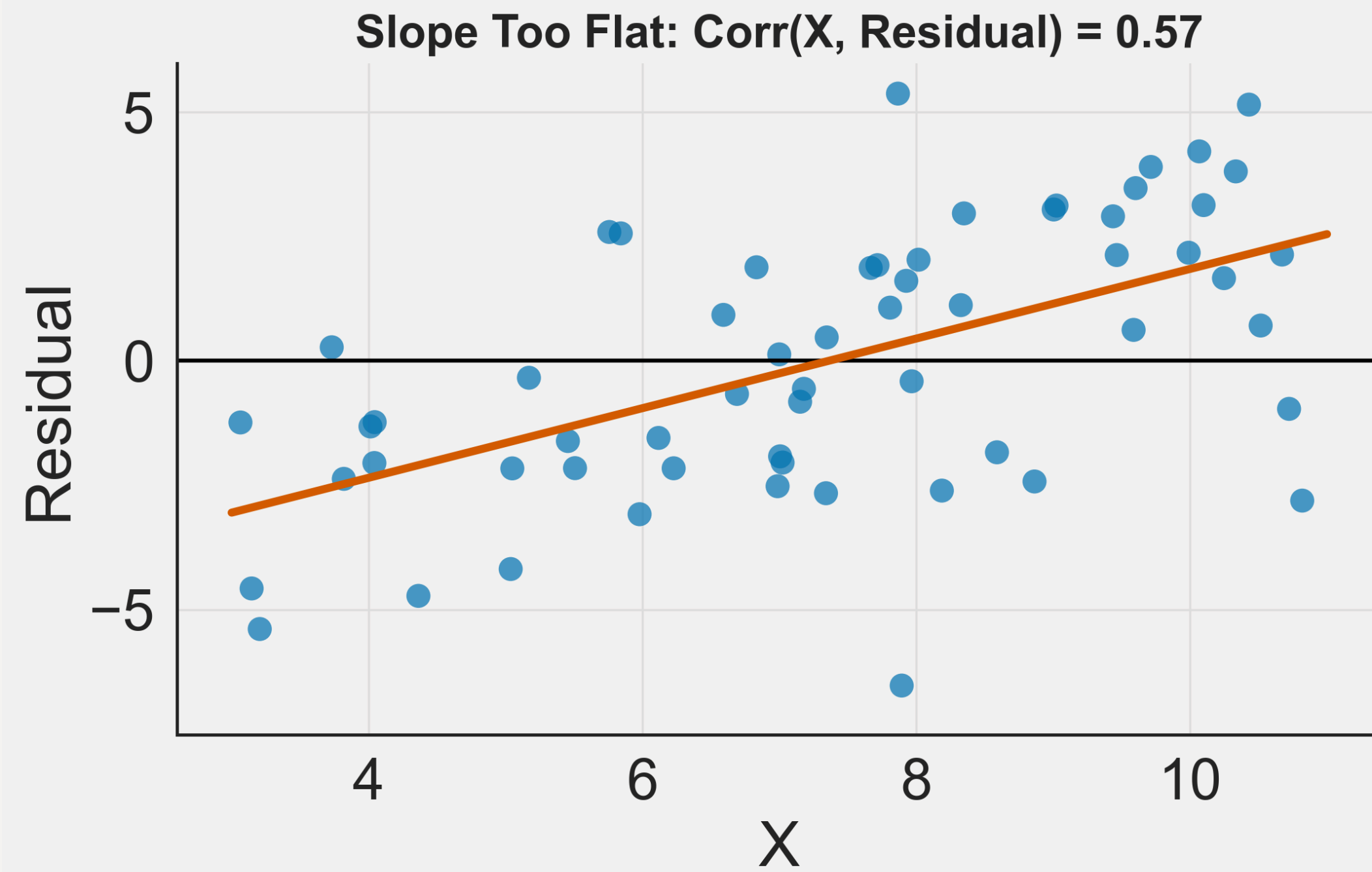
$$\sum_{i=1}^n X_i \left(Y_i - \hat{\alpha} - \hat{\beta} X_i \right) = 0$$

The **blue part** is the residual $\hat{\epsilon}_i$.

- The entire equation is a dot product.
- thus the residuals are unrelated to X .

In plain English: once we've drawn the best line, there's no correlation between the errors and X .

A Pattern Means the Slope Is Off



Step 3: Substitute in $\hat{\alpha}$

We have one equation with two unknowns. But we already solved for $\hat{\alpha}$:

$$\hat{\alpha} = \bar{Y} - \hat{\beta}\bar{X}$$

Substitute it in:

$$\sum_{i=1}^n X_i \left(Y_i - \left(\bar{Y} - \hat{\beta}\bar{X} \right) - \hat{\beta}X_i \right) = 0$$

Step 4: Simplify Inside the Parentheses

Distribute the minus sign:

$$Y_i - \bar{Y} + \hat{\beta}\bar{X} - \hat{\beta}X_i$$

Group the Y terms and the $\hat{\beta}$ terms:

$$= (Y_i - \bar{Y}) + \hat{\beta}\bar{X} - \hat{\beta}X_i$$

Factor $-\hat{\beta}$ out of the last two terms:

$$= (Y_i - \bar{Y}) - \hat{\beta}(X_i - \bar{X})$$

Step 5: Put It Back and Distribute

$$\sum_{i=1}^n X_i \left[(Y_i - \bar{Y}) - \hat{\beta} (X_i - \bar{X}) \right] = 0$$

Distribute X_i and split the sum:

$$\sum_{i=1}^n X_i (Y_i - \bar{Y}) - \hat{\beta} \sum_{i=1}^n X_i (X_i - \bar{X}) = 0$$

Step 6: Solve for $\hat{\beta}$

Move the second term to the other side:

$$\hat{\beta} \sum_{i=1}^n X_i (X_i - \bar{X}) = \sum_{i=1}^n X_i (Y_i - \bar{Y})$$

Divide both sides by $\sum X_i (X_i - \bar{X})$:

$$\hat{\beta} = \frac{\sum_{i=1}^n X_i (Y_i - \bar{Y})}{\sum_{i=1}^n X_i (X_i - \bar{X})}$$

A Trick: Deviations Sum to Zero

$$\sum_{i=1}^n (X_i - \bar{X}) = \sum_{i=1}^n X_i - \sum_{i=1}^n \bar{X} = n\bar{X} - n\bar{X} = 0$$

The same is true for Y : $\sum_{i=1}^n (Y_i - \bar{Y}) = 0$.

In plain English: the mean is the balance point. Distances above it cancel distances below it.

Applying the Trick: The Numerator

We want to show $\sum (X_i - \bar{X})(Y_i - \bar{Y}) = \sum X_i(Y_i - \bar{Y})$.

Distribute $(Y_i - \bar{Y})$ over $(X_i - \bar{X})$:

$$\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) = \sum_{i=1}^n X_i(Y_i - \bar{Y}) - \bar{X} \sum_{i=1}^n (Y_i - \bar{Y})$$

The last sum is zero, so the extra term vanishes:

$$= \sum_{i=1}^n X_i(Y_i - \bar{Y}) - \bar{X}(0) = \sum_{i=1}^n X_i(Y_i - \bar{Y})$$

Applying the Trick: The Denominator

$$\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n X_i (X_i - \bar{X}) - \bar{X} \sum_{i=1}^n (X_i - \bar{X})$$

The last sum is zero:

$$\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n X_i (X_i - \bar{X})$$

The Slope, in Deviation Form

$$\hat{\beta} = \frac{\sum_{i=1}^n X_i(Y_i - \bar{Y})}{\sum_{i=1}^n X_i(X_i - \bar{X})} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

The Slope, in Deviation Form

Multiply the top and bottom by $\frac{1}{n-1}$.

$$\hat{\beta} = \frac{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}$$

The Slope Is Covariance Over Variance

$$\hat{\beta} = \frac{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}$$

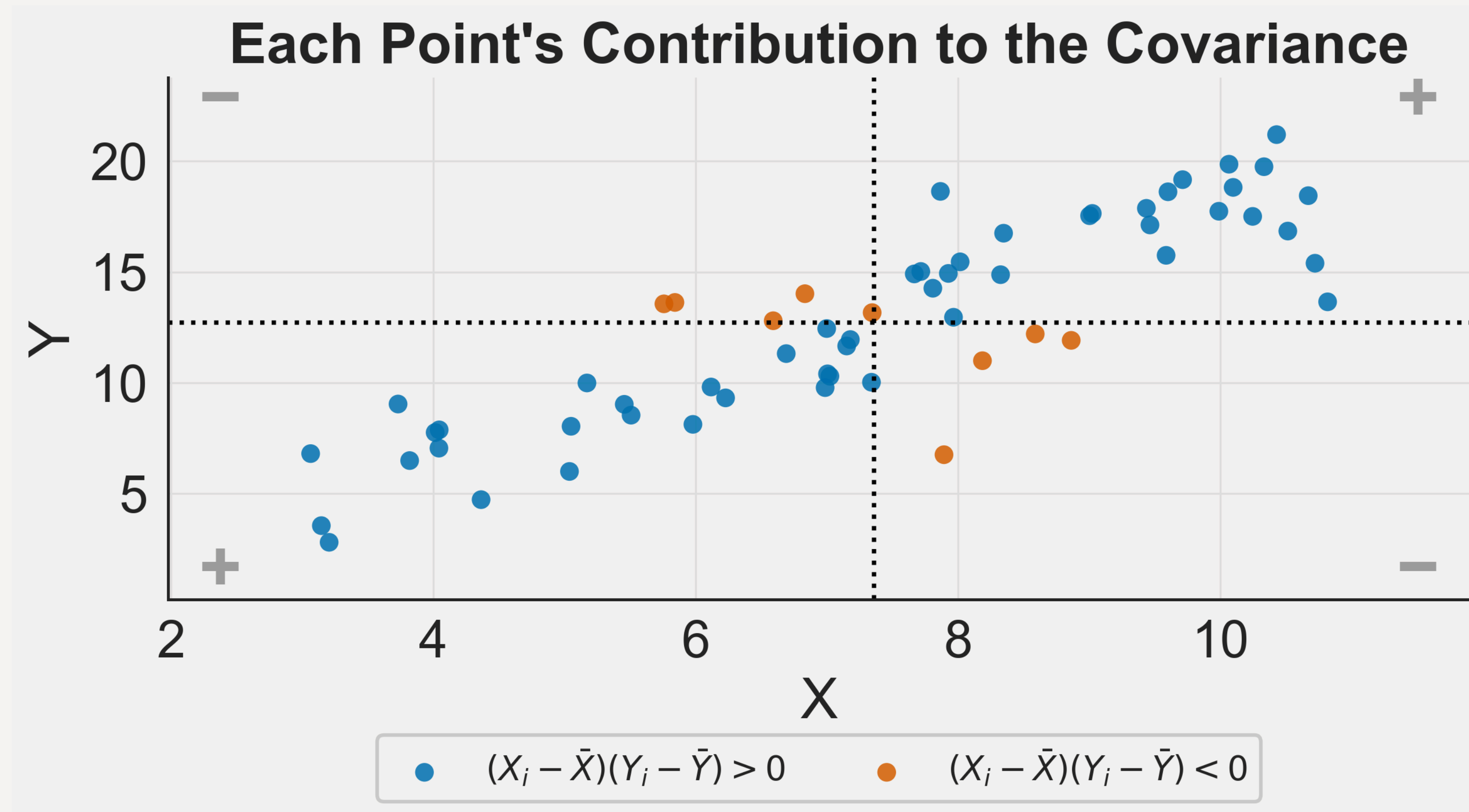
- **Numerator**: the sample covariance, s_{XY}
- **Denominator**: the sample variance of X , s_X^2

$$\hat{\beta} = \frac{s_{XY}}{s_X^2} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

Interpreting the Slope

The Numerator: Covariance

$$s_{XY} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$$



The Denominator: Variance

$$s_X^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

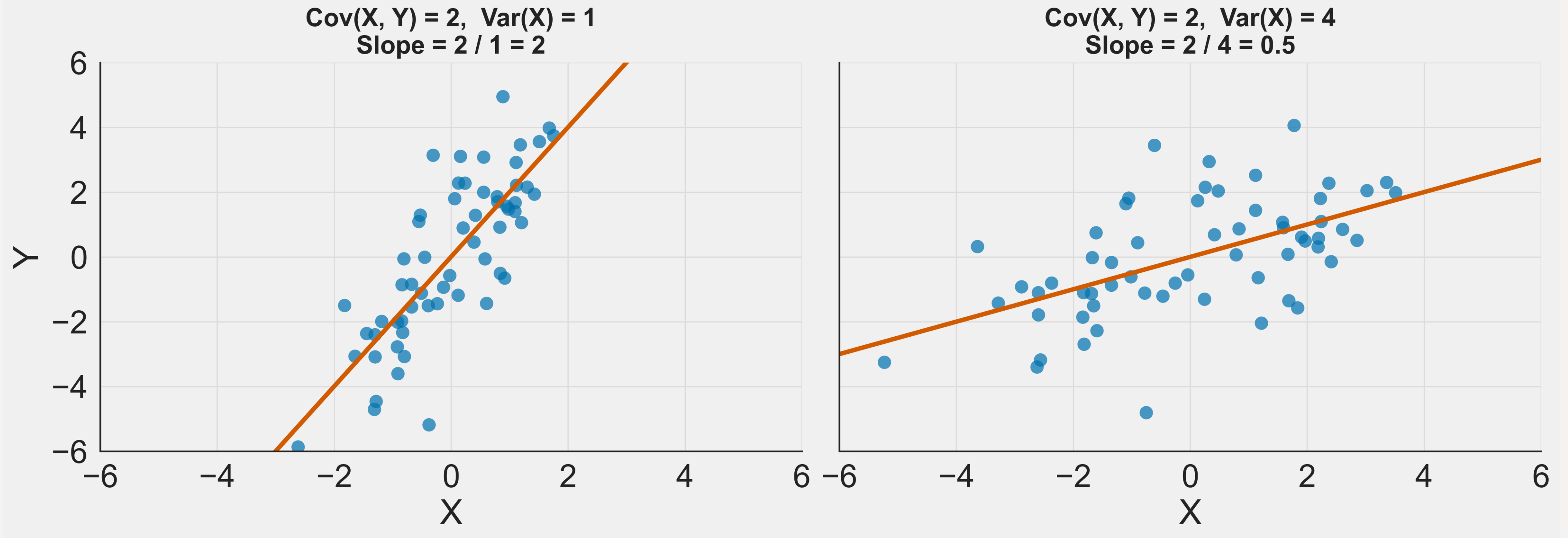
Why divide the covariance by the variance of X ?

Units. Covariance is in units of $X \times Y$. Variance of X is in units of X^2 .

$$\frac{\text{units of } X \times Y}{\text{units of } X^2} = \frac{\text{units of } Y}{\text{units of } X}$$

A slope is “change in Y per unit of X ”. Dividing by the variance of X puts it in the right units.

Same Covariance, Different Slope



Slope, Covariance, and Variance

- **Covariance:** how much X and Y move together
- **Variance of X :** how much X moves on its own
- **Slope:** how much Y moves along with each unit of movement in X

Slope and Correlation

Recall the correlation coefficient: $r_{XY} = \frac{S_{XY}}{S_X S_Y}$, so

$$S_{XY} = r_{XY} S_X S_Y.$$

Substitute into the slope formula:

$$\hat{\beta} = \frac{S_{XY}}{s_X^2} = \frac{r_{XY} S_X S_Y}{s_X^2} = r_{XY} \frac{S_Y}{S_X}$$

The slope is the correlation, rescaled by the ratio of standard deviations.

The OLS Estimates

$$\hat{\beta} = \frac{s_{XY}}{s_X^2}$$

$$\hat{\alpha} = \bar{Y} - \hat{\beta}\bar{X}$$

Summary

- OLS chooses the intercept and slope that minimize the sum of squared errors
- We find the minimum by taking the derivative with respect to each parameter and setting it to zero: at the bottom of the bowl, the slope is flat
- The intercept condition says residuals sum to zero, which forces the line through $(\bar{X}, \bar{Y})$: $\hat{\alpha} = \bar{Y} - \hat{\beta}\bar{X}$
- The slope condition says residuals are unrelated to X . Solving it gives $\hat{\beta} = \frac{s_{XY}}{s_X^2}$
- The slope is how much X and Y move together, scaled by how much X moves on its own

OLS in Matrix Form

Regression in Matrix Form

$$Y = X\beta + \varepsilon$$

- Y : $n \times 1$ column of outcomes
- X : $n \times (k + 1)$ design matrix
- β : $(k + 1) \times 1$ column of coefficients
- ε : $n \times 1$ column of errors

Fitted Values and Residuals

$$\hat{\mathbf{Y}} = \mathbf{X} \hat{\boldsymbol{\beta}}$$

$$\hat{\boldsymbol{\varepsilon}} = \mathbf{Y} - \hat{\mathbf{Y}} = \mathbf{Y} - \mathbf{X} \hat{\boldsymbol{\beta}}$$

SSE in Matrix Form

$$\hat{\boldsymbol{\varepsilon}} = \mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}$$

$$SSE(\hat{\boldsymbol{\beta}}) = \sum_{i=1}^n \hat{\varepsilon}_i^2 = \hat{\boldsymbol{\varepsilon}}^\top \hat{\boldsymbol{\varepsilon}}$$

$$SSE(\hat{\boldsymbol{\beta}}) = \left(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}} \right)^\top \left(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}} \right)$$

Expanding SSE

Step 1: Distribute the Transpose

A transposed difference distributes similarly to a squared difference [e.g. $(a - b)^2 = a^2 - 2ab + b^2$]:

$$\begin{aligned} (\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}})^\top (\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}) &= \mathbf{Y}^\top \mathbf{Y} - \mathbf{Y}^\top \mathbf{X} \hat{\boldsymbol{\beta}} \\ &\quad - \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{Y} + \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} \end{aligned}$$

Expanding SSE

Step 2: The Middle Terms Are the Same Number

$\mathbf{Y}^\top \mathbf{X} \hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{Y}$ are scalars and each is the transpose of the other:

$$\left(\mathbf{Y}^\top \mathbf{X} \hat{\boldsymbol{\beta}} \right)^\top = \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{Y}$$

Because a single number is its own transpose, these two terms are equal and we can combine them:

$$SSE(\hat{\boldsymbol{\beta}}) = \mathbf{Y}^\top \mathbf{Y} - 2 \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{Y} + \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}}$$

Derivatives With Respect to a Vector

The “derivative” of a vector is the **gradient**. It contains the partial derivative of each element.

$$\nabla_{\hat{\beta}} SSE = \begin{pmatrix} \frac{\partial SSE}{\partial \hat{\beta}_0} \\ \frac{\partial SSE}{\partial \hat{\beta}_1} \\ \vdots \\ \frac{\partial SSE}{\partial \hat{\beta}_k} \end{pmatrix}$$

Setting the gradient to **0** implies the rate of change is zero in every direction.

The gradient of SSE

$$SSE(\hat{\beta}) = \mathbf{Y}^\top \mathbf{Y} - 2 \hat{\beta}^\top (\mathbf{X}^\top \mathbf{Y}) + \hat{\beta}^\top (\mathbf{X}^\top \mathbf{X}) \hat{\beta}$$

- $\mathbf{Y}^\top \mathbf{Y}$: constant term, derivative is $\mathbf{0}$
- **Blue term**: reduces to $-2\mathbf{X}^\top \mathbf{Y}$
- **Orange term**: reduces to $2\mathbf{X}^\top \mathbf{X} \hat{\beta}$

The Gradient of SSE

$$\nabla_{\hat{\beta}} SSE = -2\mathbf{X}^\top \mathbf{Y} + 2\mathbf{X}^\top \mathbf{X} \hat{\beta}$$

$$-2\mathbf{X}^\top \mathbf{Y} + 2\mathbf{X}^\top \mathbf{X} \hat{\beta} = \mathbf{0}$$

Divide by 2 and rearrange:

$$\boxed{\mathbf{X}^\top \mathbf{X} \hat{\beta} = \mathbf{X}^\top \mathbf{Y}}$$

These are the **normal equations**.

Why are the Normal Equations “Normal”?

In geometry, normal means perpendicular.

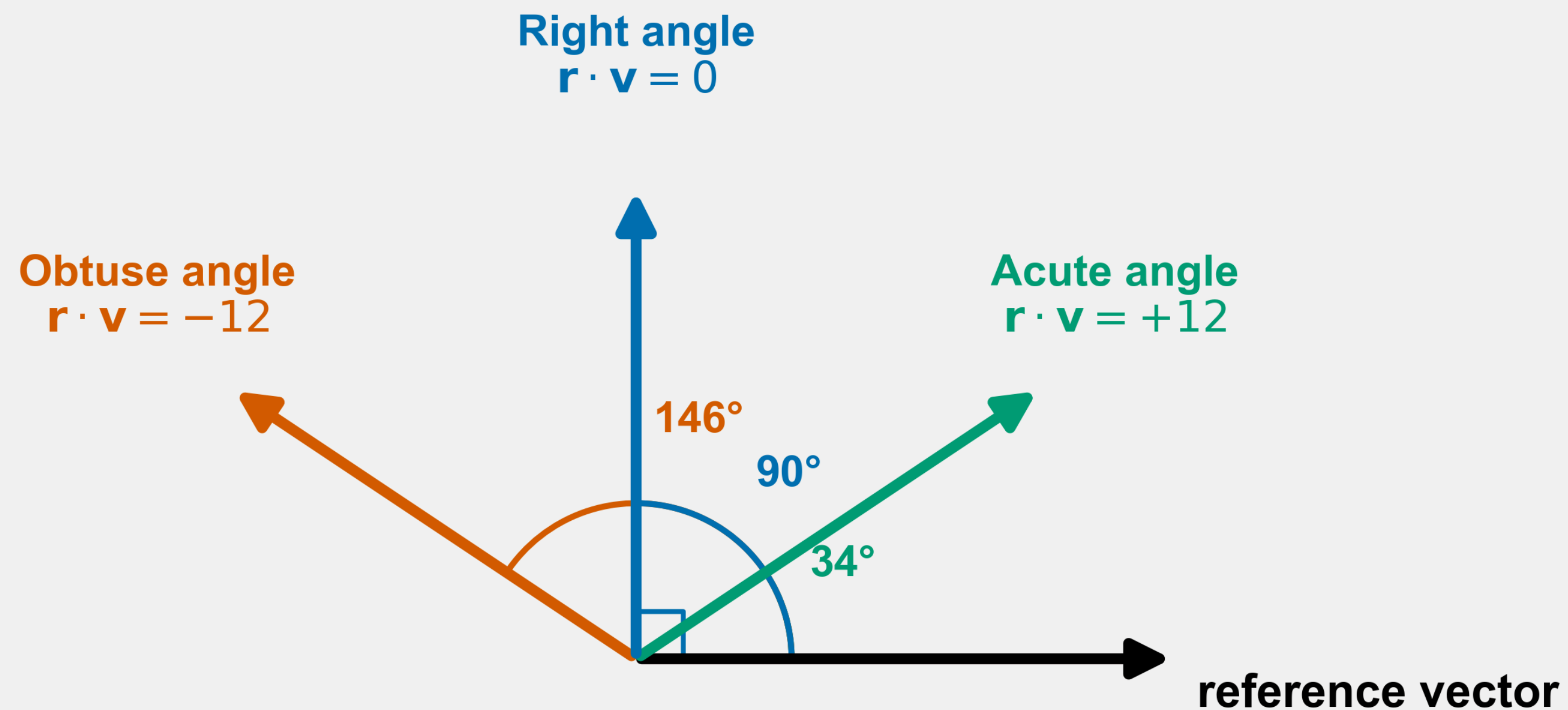
Rearranged, $\mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{Y}$ says:

$$\mathbf{X}^\top (\mathbf{Y} - \mathbf{X} \hat{\boldsymbol{\beta}}) = \mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{0}$$

- Each row of $\mathbf{X}^\top$ is a column of $\mathbf{X}$. So this says: the dot product of every X and the residual vector is zero.
- Thus every X is uncorrelated with the residuals.
- In statistics, when you hear perpendicular or orthogonal, think uncorrelated.

Why Zero Dot Product Means Perpendicular

The Sign of the Dot Product Tells You the Angle



Solving for $\hat{\beta}$

$$\mathbf{X}^\top \mathbf{X} \hat{\beta} = \mathbf{X}^\top \mathbf{Y}$$

Multiply by the inverse of $\mathbf{X}^\top \mathbf{X}$ to isolate $\hat{\beta}$.

$$(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} \hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$$

$(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}$ is the identity matrix, so it disappears:

$$\boxed{\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}}$$

The Geometry of OLS

A Different Way to See OLS: Vectors, Not Points

All lecture, we've plotted n points in a small (X, Y) scatterplot. Now flip the picture: think of $\mathbf{Y}$ itself as a **single vector**, living in n -dimensional space — one axis per *observation*, not per variable. For $n = 3$: $\mathbf{Y} = (Y_1, Y_2, Y_3)^\top$ is *one point* in 3D space, not three points on a line.

The Column Space of X

Every choice of $\hat{\beta}$ produces a fitted-value vector $X\hat{\beta}$ — some combination of X 's columns.

The set of every vector reachable this way is called the **column space of X** : a flat subspace (a line, plane, or hyperplane, depending on k) passing through the origin.

No matter what $\hat{\beta}$ we pick, $\hat{Y} = X\hat{\beta}$ is stuck **inside** this column space. We can't fit any vector outside it.

OLS Picks the Closest Point in the Column Space

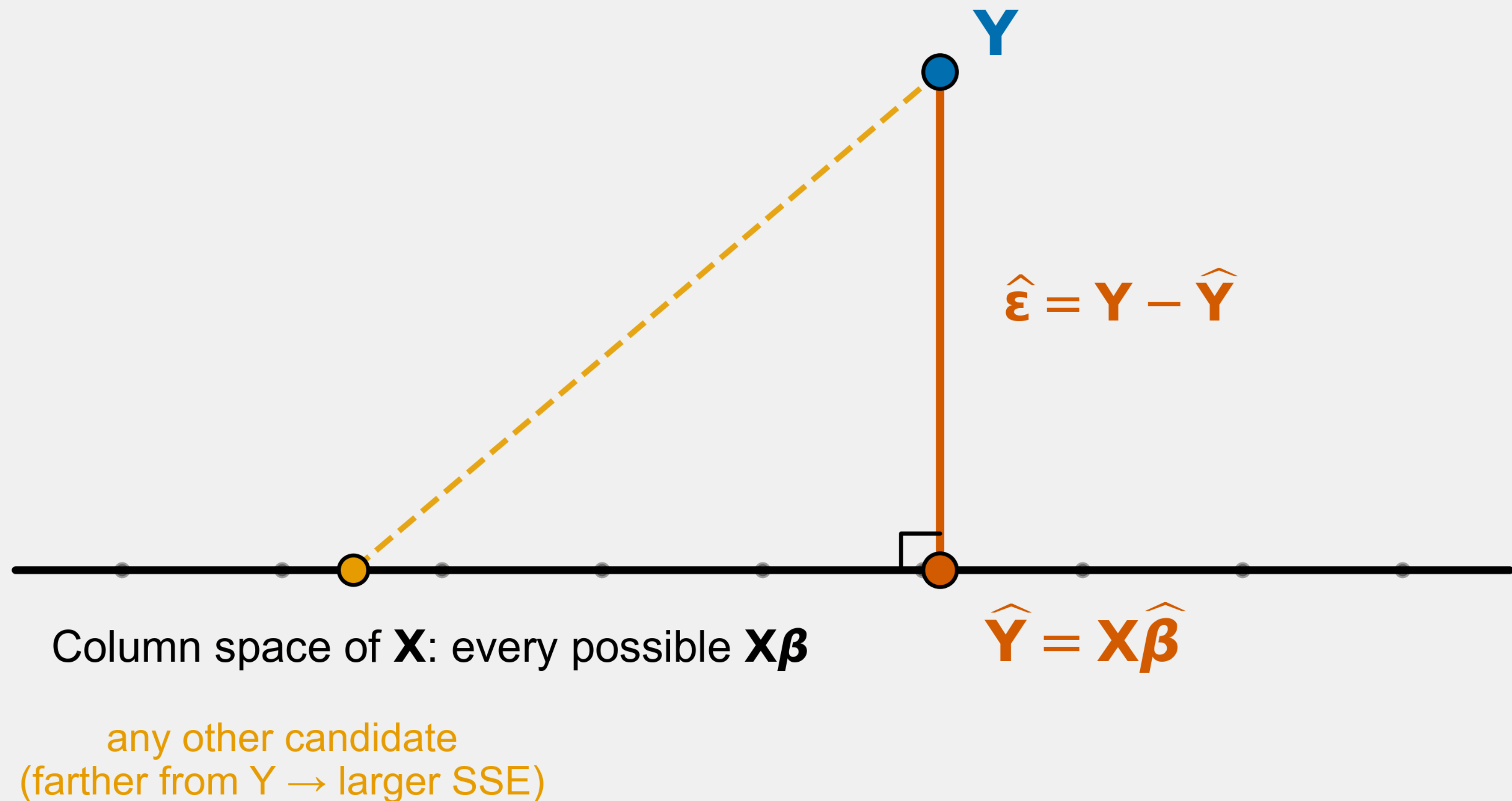
$$SSE = \hat{\boldsymbol{\varepsilon}}^\top \hat{\boldsymbol{\varepsilon}} = \|\mathbf{Y} - \hat{\mathbf{Y}}\|^2$$

SSE is the squared *distance* between $\mathbf{Y}$ and $\hat{\mathbf{Y}}$.

Minimizing SSE = finding the point in the column space of $\mathbf{X}$ that is closest to $\mathbf{Y}$.

Min(SSE) Visualized

OLS Finds the Closest Point to Y Inside the Column Space of X



Why Perpendicular? Back to the Normal Equations

The shortest path from a point to a flat surface is always **perpendicular** to that surface.

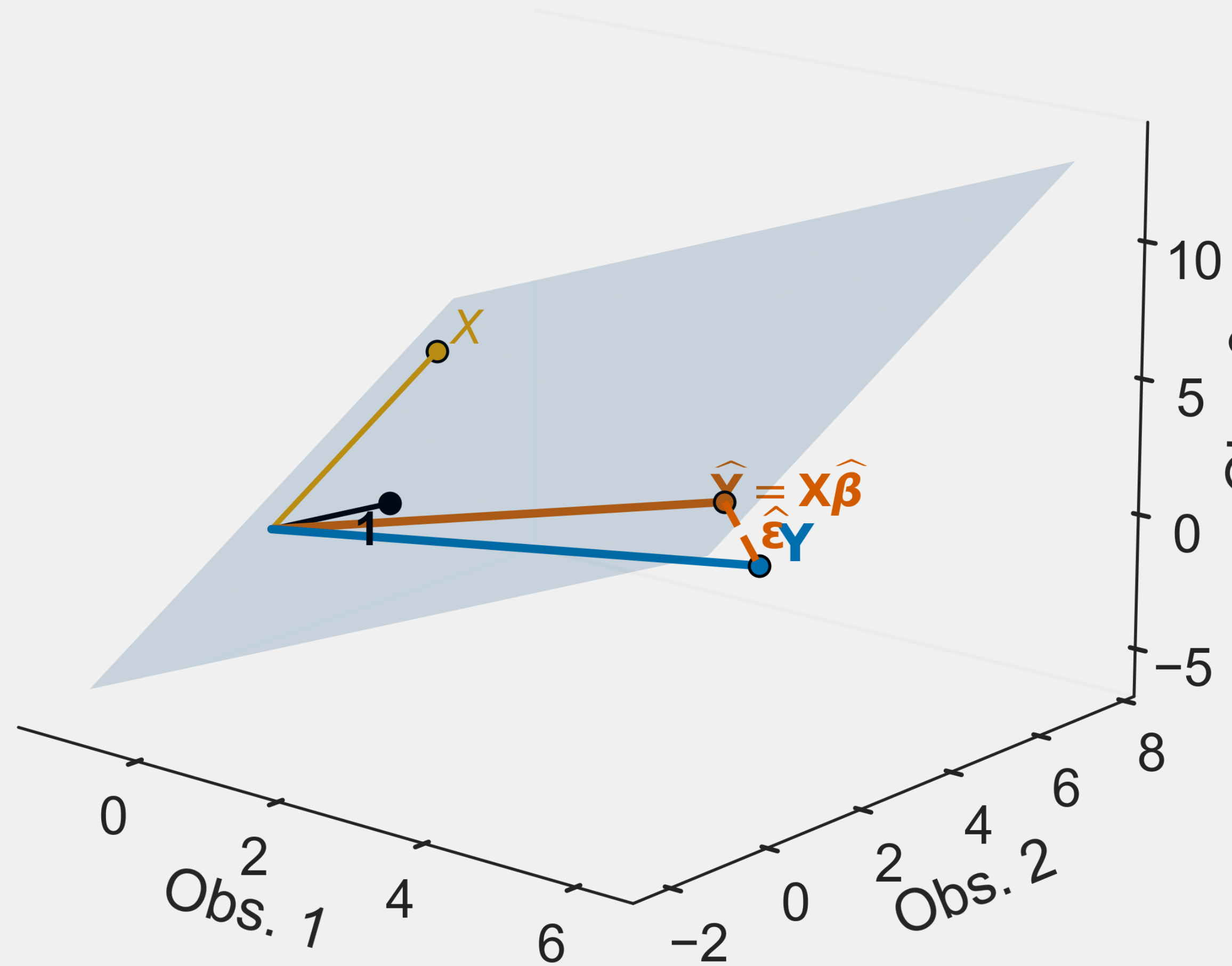
That's exactly what the normal equations told us algebraically:

$$\mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{0} \quad \Longleftrightarrow \quad \hat{\boldsymbol{\varepsilon}} \text{ is orthogonal to every column of } \mathbf{X}$$

Two completely different derivations — calculus, and “shortest distance to a plane” — arrive at the identical condition.

Orthogonality Visualized

Column Space of X (the shaded plane), Y , and $\hat{\epsilon} \perp$ the Plane



The Hat Matrix

Combine the fitted-value equation with the OLS solution:

$$\hat{\mathbf{Y}} = \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y} = \mathbf{H} \mathbf{Y}, \quad \mathbf{H} \equiv \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$$

$\mathbf{H}$ is called the **hat matrix**: it's the matrix that puts the “hat” on $\mathbf{Y}$.

Summary: The Algebra

- Matrix algebra gives one derivation that works for any number of predictors, instead of redoing calculus for each new parameter
- $SSE(\hat{\beta}) = \mathbf{Y}^\top \mathbf{Y} - 2\hat{\beta}^\top \mathbf{X}^\top \mathbf{Y} + \hat{\beta}^\top \mathbf{X}^\top \mathbf{X} \hat{\beta}$: a bowl in $(k + 1)$ dimensions
- Setting the gradient to zero gives the **normal equations**:
 $\mathbf{X}^\top \mathbf{X} \hat{\beta} = \mathbf{X}^\top \mathbf{Y}$, equivalent to $\mathbf{X}^\top \hat{\varepsilon} = \mathbf{0}$ — every column of $\mathbf{X}$ uncorrelated with the residuals, generalizing both scalar conditions at once

Summary: The Solution and Its Geometry

- Closed-form solution: $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$, valid whenever $\mathbf{X}$ has full column rank
- Geometrically: OLS finds the point in the column space of $\mathbf{X}$ closest to $\mathbf{Y}$; the residual is the perpendicular leftover piece
- Two completely different routes — calculus, and “closest point in a subspace” — land on the exact same $\hat{\beta}$

Key Definitions

Key Definitions

- **Ordinary Least Squares (OLS):** the method that chooses the coefficients that minimize the sum of squared errors
- **Sum of squared errors (SSE):** $\sum_{i=1}^n (Y_i - \hat{Y}_i)^2$; how far a line is from the data, overall
- **Derivative:** the slope of a function at a point; how much the output changes for a tiny nudge in the input
- **Partial derivative:** the derivative with respect to one variable, treating all others as constants
- **First-order condition:** setting a derivative equal to zero to find where a function is flat

Key Definitions

- **Sum rule:** the derivative of a sum is the sum of the derivatives
- **Chain rule:** for nested functions, the derivative is (derivative of the outside) $\times$ (derivative of the inside)
- **Deviation from the mean:** $X_i - \bar{X}$; deviations always sum to zero
- **Sample covariance:** $s_{XY} = \frac{1}{n-1} \sum (X_i - \bar{X})(Y_i - \bar{Y})$
- **Sample variance:** $s_X^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$

Key Definitions

- OLS slope: $\hat{\beta} = \frac{s_{XY}}{s_X^2}$
- OLS intercept: $\hat{\alpha} = \bar{Y} - \hat{\beta}\bar{X}$
- **Properties of the OLS line:** passes through $(\bar{X}, \bar{Y})$; residuals sum to zero; residuals are uncorrelated with X

Key Definitions

- **Gradient** ($\nabla_{\hat{\beta}} SSE$): the vector of partial derivatives of SSE with respect to each entry of $\hat{\beta}$; zero at the bottom of the bowl
- **Quadratic form**: an expression like $\mathbf{x}^\top \mathbf{A} \mathbf{x}$; SSE is a constant, plus a linear form, plus a quadratic form in $\hat{\beta}$
- **Normal equations**: $\mathbf{X}^\top \mathbf{X} \hat{\beta} = \mathbf{X}^\top \mathbf{Y}$, the matrix first-order condition; equivalent to $\mathbf{X}^\top \hat{\epsilon} = \mathbf{0}$
- **Full column rank**: no column of $\mathbf{X}$ is an exact linear combination of the others; needed for $(\mathbf{X}^\top \mathbf{X})^{-1}$ to exist

Key Definitions

- **OLS solution (matrix form):** $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$
- **Column space of $\mathbf{X}$:** every vector reachable as $\mathbf{X}\beta$ for some β ; the flat subspace that $\hat{\mathbf{Y}}$ is confined to
- **Orthogonal projection:** finding the closest point to $\mathbf{Y}$ within a subspace; geometric description of what OLS does
- **Hat matrix ($\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$):** the matrix satisfying $\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$